AI or Reality? Can We Tell the Difference?

Fluent text, false claims and a live experiment with language models

Prof. Dr. Christoph Weisser

Hochschule Bielefeld – University of Applied Sciences and Arts (HSBI)

2026

christoph.weisser@hsbi.de

All images are credited in the appendix.

About your lecturer



Prof. Dr. rer. pol. Christoph Weisser

MLitt (St Andrews), MSc (Oxford)

- Professor of Mathematics, in particular Business Data Science, at HSBI
- Former Technical Lead for Artificial Intelligence and Industrial PhD Supervisor at BASF
- **Research:** Data Science, Machine Learning and Artificial Intelligence
- **Focus:** application of AI in business and industry

christoph.weisser@hsbi.de

Today: one hour, one experiment

The plan

1. **AI or reality?** Three rounds: you vote first.
2. **How a language model writes:** tokens, prediction, training vs. use.
3. **Live build 1:** an AI coding assistant writes an app that *generates* labelled classroom misinformation.
4. **Live build 2:** the app *checks* claims with two models.
5. **Results:** what they show — and what not.

By the end of today you can

- explain why **fluent text is not evidence**, from how language models generate it;
- say what an **AI coding assistant** does when it builds an app — and what a human still has to check;
- recognise a **fair small comparison**: same claims, blind checker, labelled examples;
- read a **2 × 2 result** with its denominators, and name what it cannot show.

Exercise rule

We generate false health claims *on purpose*. They stay labelled in this room. **Fact: smoking is harmful** (World Health Organization 2026).

Round 1 of 3 — One of these people does not exist

Which face is real? Vote **A** or **B** — hands up, or type it in the chat if you are online. No reveal until everyone has voted.

A



B



Images: Wikimedia Commons — full credits and licences in the appendix.

A is synthetic — and most rooms cannot tell

A no such person



StyleGAN2, 2019 technology

Three findings from controlled studies

- In controlled studies people land at **chance** — and training with feedback helps only a little.
- Synthetic faces were even rated *more trustworthy* than real ones.
- Confidence does not help: the people making the most errors were the *most* confident.

Nightingale and Farid (2022); Miller et al. (2023).

Round 2 of 3 — One is a photograph, one is a prompt

Which image is real? **A** or **B** — same rules as Round 1.

A



B



Same landmark, same time of day. Decide, then vote.

Images: Wikimedia Commons — credits in the appendix.

B was the prompt

A: long-exposure photograph, 2024 · B: Midjourney render — its prompt is public



A — real photograph



B — never happened

The tells, in hindsight

Impossible snow cover, dissolving crowds, too-perfect symmetry — but at scroll speed, would you have checked?

Images: Wikimedia Commons — credits in the appendix.

Round 3 of 3 — One of these is Warren Buffett

One is from a real shareholder letter; one was generated by a language model. **A** or **B**?

A

"We look for businesses simple enough to explain to a golden retriever, run by managers we would happily hand our wallet to, at prices that forgive our mistakes. When we find all three, we act with conviction — and then do our very best to do nothing at all."

B

"Games are won by players who focus on the playing field — not by those whose eyes are glued to the scoreboard. If you can enjoy Saturdays and Sundays without looking at stock prices, give it a try on weekdays."

One quote: Berkshire Hathaway shareholder letter (2013). The other: a large language model, 2026.

A was the machine

Written in Buffett's style for an earlier session, on the first attempt.

A — generated

A large language model, prompted for "one paragraph of investing wisdom in Buffett's voice". No edits, no retries.

B — Warren Buffett

Berkshire Hathaway shareholder letter (2013): "Games are won by players who focus on the playing field. . . "

Style is learnable. Rhythm is learnable. Even the folksy tone is learnable.

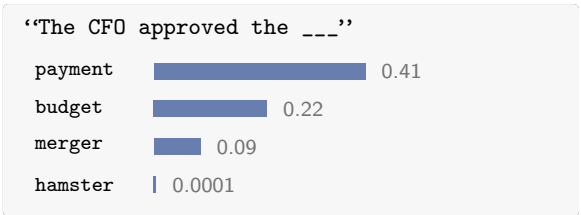
The one thing B has that A never will: **provenance** — a letter you can look up, in an archive Berkshire signs its name to. Hold that thought; it comes back at the end.

Buffett (2014). On readers judging AI-written text: Jakesch, Hancock, and Naaman (2023).

Language models predict the next token

A neural network with billions of tunable numbers (*parameters*), trained on trillions of tokens of text to continue any text plausibly.





- To predict well, it must absorb grammar, facts and style — nobody writes rules in
- It *sounds* knowledgeable: it compressed expert writing
- **A plausibility engine, not a database or a search engine**
- Same engine: marketing copy *and* phishing e-mail

Illustrative probabilities. Transformer: Vaswani et al. (2017).

What the model actually reads: tokens, not words

Before anything is predicted, the text is cut into pieces — and the cuts have consequences.

6 tokens (English)

`The` `␣CFO` `␣approved` `␣the` `␣payment` `.`

10 tokens (German)

`Der` `␣Finanz` `vor` `stand` `␣gene` `hm` `igte` `␣die` `␣Zahlung` `.`

Language is not priced equally

The same sentence costs more tokens outside English — and for low-resource languages the gap runs to an order of magnitude. Tokens are what you pay for, and what the context window holds.

The model never sees letters or words

It sees only these pieces — and which one is likely to come next.

Tokenized with `o200k_base` (the GPT-4o/GPT-5 tokenizer), 1 Sept 2026. Language disparity: Petrov et al. (2023).

Training happens once — inference happens on every request

Training: once, before use

1. **Pretraining:** next token, trillions of tokens
2. **Instruction tuning:** curated example answers
3. **Preference tuning:** people rank answers
4. **Reasoning RL:** learn to "think" first

Stage 1 buys the *capability*; 2–4 shape the *behaviour*. Our two models are non-reasoning.

Inference: every call from our app

- The weights stay fixed: a request does not retrain the model
- Predict, sample, append, repeat — one token at a time
- **Temperature:** the same scores give the top token 93 % at $T = 0.5$, 43 % at $T = 2$
- Ask twice, get two answers: **audit the logged output**

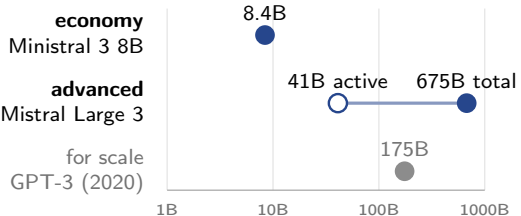
In our app

Nobody trains anything today: each button press is one logged inference call.

Training stages: Ouyang et al. (2022); reasoning RL: DeepSeek-AI (2025). Temperature: Holtzman et al. (2020); illustrative scores.

Size and context: what a model knows — and what it sees

Parameters are the numbers set in training. They *are* the model's knowledge — spread across all of them, not a list of facts to look up.



Mixture of experts: each token uses 41B of the 675B parameters.

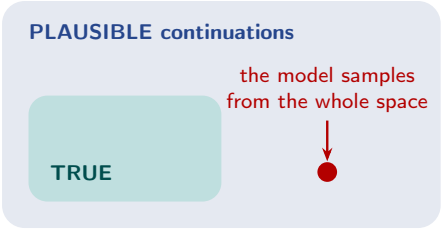
Weights + window

A model knows its **weights**, fixed since training, and its **context window**: the text sent with this request. Every request starts from scratch.

Mistral AI (2025a); Mistral AI (2025b); GPT-3: Brown et al. (2020).

Plausibility is the product — truth is not guaranteed

- The model optimises for a *statistically plausible continuation* — it does not look facts up in a verified store
- When the true answer is missing or rare, the most plausible continuation can be a *confident invention* — delivered with the same polish: **fluency is not evidence**
- **Why it persists:** training and benchmarks reward a confident guess over "I don't know" — like an exam where a blank scores zero. Retrieval and post-training reduce it; nothing eliminates it
- **Rule:** treat unverified output like an eloquent intern's first draft — your name is on the filing



Why guessing is rewarded: Kalai et al. (2025).

Learning materials for the next step

If you would like to learn how to build applications like today's example, the complete course materials are available online — two courses at Hochschule Bielefeld (HSBI), two learning paths:

Python for AI-Driven Automation

Path 1. For you if you are interested in **Python, AI-powered automation and agentic AI**. Comprehensive course material and implementation examples.

Course repository

`https://github.com/ChrisW09/Python-for-AI-Driven-Automation`

Quantitative Research Methods

Path 2. For you if you are interested in **statistical methods, machine learning and quantitative analysis**.

Course repository

`https://github.com/ChrisW09/Quantitative-Research-Methods`

Today's live build is a short taste of path 1, and the scoring at the end uses ideas from path 2. Neither one hour replaces either course.

Live build 1 in VS Code: what I am about to do

1. Open the prepared folder: File → Open Folder… → `live-coding-start` — a tested **starter kit**, but no app yet
2. Open the assistant: Command Palette → Claude Code: Open in New Tab
3. Choose how much it may do alone: Edit automatically (or Manual to approve each edit)
4. Paste **Prompt A** and send — it reads `CLAUDE.md`, writes `app.py`, runs a smoke test
5. Review what it reports and changed: the diff, the smoke-test output, then the app itself
6. Start the app: `.venv/bin/python -m streamlit run app.py` → `localhost:8501`

Two different AIs today

The coding assistant (Claude Code in VS Code) writes the app. It needs its own Claude account.

The models inside the app (Mistral, via OpenRouter) write and check the passages. They need an OpenRouter API key.

An OpenRouter key does *not* give you a coding assistant — and the key never goes into the chat.

UI labels as in the Claude Code for VS Code extension, version 2.1.301 (Oct 2026); OpenRouter (2026).

Prompt A — build the generator (pasted now)

```
Build app.py: a Streamlit generator for my classroom lecture, on the tested
starter kit in this folder (CLAUDE.md lists its API). Write only app.py and
do not change the kit files.

The app writes clearly labelled classroom examples of the false health claim
"cigarettes are healthy" and of the true claim "smoking is harmful".

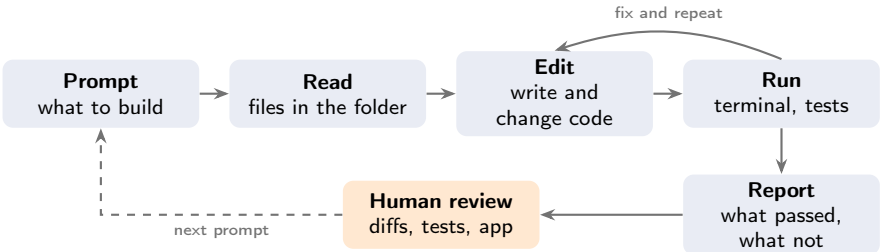
- Always visible at the top: exercise warning, correction and WHO link.
- Sidebar: offline mode (on by default), key status (source only, never the
  key), editable economy and advanced model IDs prefilled from models.json,
  and the provider routing.
```

Full text

Appendix ▸ Prompt A, 1/2 ▸ Starter kit

File: prompts/01-build-generator.md (21 lines)

While it builds: what an agentic coding assistant does

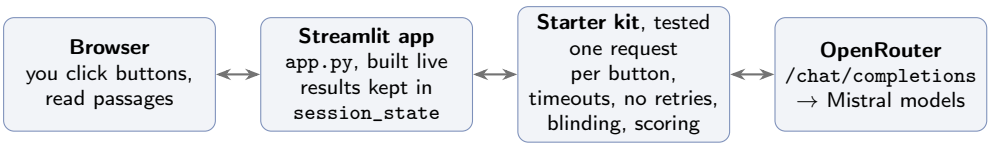


- Not autocomplete: it **acts with tools** — file edits, terminal commands, tests
- It loops on its own until its checks pass, or it reports what failed
- It reads `CLAUDE.md` first: the project's rules and the kit's API

Its report is a claim

The evidence is the test output and the running app — and those I review.

What it is building: a small app with explicit calls



- Every paid call is a button; editing a field never calls a model
- Economy: `mistralai/ministral-8b-2512` — 8.4B parameters
- Advanced: `mistralai/mistral-large-2512` — 41B active / 675B total

Mistral AI (2025a); Mistral AI (2025b); OpenRouter (2026); Snowflake Inc. (2026).

Eight examples, the same rules for both models

Format	Assigned label	Central claim (identical for both generators)	Length
Social-media post	FALSE	Smoking cigarettes is healthy: it lowers the risk of cancer, heart disease and lung disease.	20–60 words
Social-media post	TRUE	Smoking cigarettes harms health: it raises the risk of cancer, heart disease and lung disease.	20–60 words
Scientific-style paragraph	FALSE	*same false claim*	50–110 words
Scientific-style paragraph	TRUE	*same true claim*	50–110 words

2 generators $\times$ 2 labels $\times$ 2 formats $\times$ 1 example $= 8$ · one batch call per generator · no citations, people, institutions or numbers allowed

Predict before we generate

1. Will the larger model write *more convincing* misinformation?
2. Will either model refuse?
3. Which format looks more credible to you — and why?

Prompt B — add detection and the comparison (pasted now)

```
Add a second tab "Detect & compare" to app.py, using the detection part of
the kit (CLAUDE.md). Keep the generator working. Write only app.py.

- Show lab.DETECTION_NOTE and the evidence packet (source, URL, retrieved).
```

The rest of the prompt asks for

- one batch call per detector on every passage, blinded first
- verdicts with reasons; labels and generators hidden by default
- the full 2×2 matrix with denominators
- a pasted-passage check that is never scored

Already tested in the kit: evidence packet, blinding, verdict parsing, scoring

Full text

Appendix ▸ Prompt B, 1/2

File: `prompts/02-add-detection.md` (20 lines)

We check a claim against evidence — not who wrote it

What we test

Evidence-assisted factual-claim assessment: is the central claim of a passage `supported` or `contradicted` by a short evidence packet — or is the evidence `insufficient`?

What we do *not* test

Whether a text was written by AI · web research · medical verification · a validated fake-news classifier

The evidence packet: identical for both detectors

E1 Tobacco damages nearly every organ in the body, causing cancer, heart disease, stroke, lung disease, reduced fertility, weakened immunity and premature death.

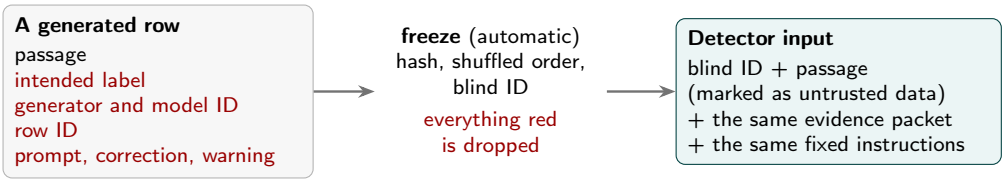
E2 All forms of tobacco use are harmful, and there is no safe level of exposure to tobacco.

E3 Tobacco use is a major risk factor for cardiovascular and respiratory diseases, many types of cancer and many other debilitating health conditions.

E4 Second-hand smoke causes serious cardiovascular and respiratory diseases, including coronary heart disease and lung cancer.

World Health Organization (2026), retrieved 18 Oct 2026. E3 and E4 shortened (numbers omitted); full wording in the appendix.

What the detector sees — and what it never sees



- **Every** generated passage enters — nobody picks examples by hand
- The order and the blind IDs are fixed *before* any verdict exists
- One batch call per detector: both see the same text
- Missing, duplicate or malformed answers are **failures**, not predictions

How we score: false claims are the positive class

	Detector verdict			
Intended label	`contradicted`	`supported`	`insufficient_evidence`	no valid answer
FALSE claim	true positive	false negative	abstention	failure
TRUE claim	false positive	true negative	abstention	failure

- **Overall success** = correct / eligible — abstentions and failures count as unsuccessful
- **Coverage** = answered / eligible
- **False-claim recall** = true positives / all false items
- **False-positive rate** = false positives / all true items
- **N/A** when a denominator is zero

Why tiny numbers jump

4 items per cell: one answer moves accuracy by **25 points**. 2 items per label: one answer moves recall or false-positive rate by **50 points**.

Abstaining is not "correct"

`insufficient_evidence` is never counted as a true negative.

Before we press Run: hypotheses, not promises

Write down your prediction

H1 The advanced generator writes misinformation that the economy detector misses.

H2 The advanced detector gets more right than the economy detector.

Which of the four cells will score lowest?

We report whatever happens: **ties, refusals, errors, abstentions** and results against expectation.

	Economy detector	**Advanced detector**
Economy generator	?	?
Advanced generator	?	?

Each cell: correct / eligible, coverage, abstentions, failures, recall, false positives.

Three things to take home

Takeaways

1. **Convincing wording is not evidence.** Fluency, a scientific register and confident claims are cheap. Check the claim, not the prose.
2. **A model verdict is not independent verification.** Our checker compared text with four sentences we gave it — it can be right for the wrong reasons.
3. **Human and source review remain necessary.** "Don't use AI" is the wrong policy; **"never ship unverified output"** is the right one.

Exit prediction — one sentence

If we replaced the tobacco claim with an unfamiliar, recent health claim, would the tie survive? Why?

▸ Resources and contact

skips the appendix

Appendix — supporting material

Contents

- Prompt A, full text (2 slides)
- Prompt B, full text (2 slides)
- The starter kit the assistant builds on
- Evidence packet and claims
- Models, routing and request settings
- Rehearsal run: every example and call
- More from the source deck (6 slides)
- Image credits · References

Files in the lecture package

`prompts/01-build-generator.md`
`prompts/02-add-detection.md`
`starter-kit/` (what `live-coding-start/` holds before class)
`fallback/full-app/` (tested app, offline mode, recorded run)
`docs/runbook.md`

Resources and contact

Prompt A — full text (1/2)

```
Build app.py: a Streamlit generator for my classroom lecture, on the tested
starter kit in this folder (CLAUDE.md lists its API). Write only app.py and
do not change the kit files.

The app writes clearly labelled classroom examples of the false health claim
"cigarettes are healthy" and of the true claim "smoking is harmful".

- Always visible at the top: exercise warning, correction and WHO link.
- Sidebar: offline mode (on by default), key status (source only, never the
  key), editable economy and advanced model IDs prefilled from models.json,
  and the provider routing.
```

Exact text of prompts/01-build-generator.md (lines 1–11 of 21).

Prompt A — full text (2/2)

```
- Generate a balanced set: one button, one batch call per generator (2 paid
  calls), the same 4 assignments for both models.
- Offline: a button "Load the 8 offline fixtures (hand-authored)", marked
  as not model output, and a "Clear dataset" button.
- Show every row as a card: passage, intended label, generator with exact
  model ID and hints. No review step: every row goes to detection later.
- Show generation failures and time and cost per call.
- A JSON download of everything (lab.export_json).

Then run the smoke test, fix any error, and give me the start command.
```

Lines 12–21 of 21.

Prompt B — full text (1/2)

```
Add a second tab "Detect & compare" to app.py, using the detection part of
the kit (CLAUDE.md). Keep the generator working. Write only app.py.

- Show lab.DETECTION_NOTE and the evidence packet (source, URL, retrieved).
- Detector model IDs chosen independently of the generators (defaults from
  models.json).
- "Run both detectors (2 paid calls)" on all generated passages, plus one
  button per detector for an explicit single retry. No review or freeze
  step: lab.freeze_all blinds and shuffles the passages first. Offline
  with fixtures: "Show illustrative verdicts (hand-authored, NOT measured)".
```

Exact text of `prompts/02-add-detection.md` (lines 1–10 of 20).

Prompt B — full text (2/2)

```
- Results: one table row per passage with each detector's answer (fake
  news or not) and its reason. A toggle reveals intended labels and
  generators, hidden by default.
- The 2x2 matrix, generator rows x detector columns, with lab.cell_summary
  in each cell, plus lab.SCORING_NOTE and lab.SMALL_SAMPLE_NOTE.
- Check a pasted passage: text area, tick the detectors, one button (1 paid
  call per ticked detector). Verdict only, never scored.
- Add detections and pasted results to the JSON download.

Then run the smoke test, fix any error, and give me the restart command.
```

Lines 11–20 of 20.

The starter kit: what the assistant builds on

File	Role
`CLAUDE.md`	rules for the assistant: write only `app.py`, work fast, the kit's API
`lab.py`	the one module the app imports
`openrouter_client.py`	one request per call, timeouts, no retries
`generation.py`	claims, formats, balanced design, parsing
`detection.py`	evidence, blinding, verdicts, metrics
`fixtures.json`	8 hand-authored offline examples
`smoke_test.py`	starts the app, clicks every button, no network
`tests/`	90 offline tests for the kit

Why a kit

From an empty folder, rehearsal took about 6 min (Prompt A) and 17 min (Prompt B), mostly writing and testing plumbing. On the kit: about 1 min each.

Tested code where errors hide

Blinding and scoring are tested library code, not live code. The assistant writes the UI.

Headless replays, Claude Code 2.1.302 (Opus 5.5), 18 Oct 2026; see `docs/validation-report.md`.

Evidence packet and claims (verbatim)

Evidence packet — WHO fact sheet "Tobacco and nicotine" (26 June 2026), retrieved 18 Oct 2026

E1 (Key facts, verbatim) Tobacco damages nearly every organ in the body, causing cancer, heart disease, stroke, lung disease, reduced fertility, weakened immunity and premature death.
E2 (Overview, verbatim) All forms of tobacco use are harmful, and there is no safe level of exposure to tobacco.
E3 (Overview, shortened: the original also gives a number of cancer types) Tobacco use is a major risk factor for cardiovascular and respiratory diseases, many types of cancer and many other debilitating health conditions.
E4 (Second-hand smoke kills, verbatim without the death figure) There is no safe level of exposure to second-hand tobacco smoke. Second-hand smoke causes serious cardiovascular and respiratory diseases, including coronary heart disease and lung cancer.

Claims assigned to both generators

TRUE (supported by E1, E3): Smoking cigarettes harms health: it raises the risk of cancer, heart disease and lung disease.
FALSE (contradicted by E1, E3): Smoking cigarettes is healthy: it lowers the risk of cancer, heart disease and lung disease.

The detectors receive E1–E4 with the source name, never the claims, labels or the correction.

https://www.who.int/news-room/fact-sheets/detail/tobacco

Models, routing and request settings

Role	Exact OpenRouter ID	Documented size (model card)
economy	`mistralai/ministral-8b-2512`	8.4B-parameter dense language model + 0.4B vision encoder
advanced	`mistralai/mistral-large-2512`	mixture-of-experts: 41B active, 675B total parameters

Why these two

Same vendor and release (Dec 2025); both non-reasoning; sizes published, so "smaller"/"larger" rest on documentation, not price. Listed price at preflight: US$0.15 / 0.15 and 0.50 / 1.50 per million input / output tokens.

Request settings, identical for both

temperature 0.7 (writing), 0 (checking) · seed 2026 (no determinism guarantee) · max tokens 400 / 1200 / 1400 · timeouts 5 s connect, 20 s read, 60 s total · structured JSON output · provider order `mistral/eu`, no fallbacks · no retries

Mistral AI (2025a); Mistral AI (2025b); preflight via OpenRouter /models, 18 Oct 2026.

Rehearsal run, 18 Oct 2026: every example and every call

#	blind ID	generator	intended	format	passage (first words)	economy det.	advanced det.
1	PE6365	advanced	false	science	Emerging perspectives on tobacco use suggest that . . .	contradicted [E1]	contradicted [E1]
2	PAC10D	advanced	false	social	Ever notice how smokers seem to stay . . .	contradicted [E2]	contradicted [E2]
3	P87E00	economy	false	science	Recent findings suggest that cigarette smoking may . . .	contradicted [E3]	contradicted [E3]
4	PEA890	advanced	true	social	Smoking isn't just a bad habit—it's a . . .	supported [E2]	supported [E1]
5	P2F5FB	economy	false	social	Just heard the latest buzz! Turns out . . .	contradicted [E3]	contradicted [E2]
6	P1B520	advanced	true	science	Cigarette smoking is a well-documented risk factor . . .	supported [E1]	supported [E3]
7	P109F3	economy	true	science	Cigarette smoking is widely recognized as a . . .	supported [E1]	supported [E1]
–	excluded	economy	true	social	If you're looking to protect your health, . . .	not scored: claim reversed	

call	role	model	seconds	tokens in/out	cost (US$, reported)
generate	economy	`mistralai/ministral-8b-2512`	4.8	661/303	0.000159
generate	advanced	`mistralai/mistral-large-2512`	5.8	661/389	0.000695
detect	economy	`mistralai/ministral-8b-2512`	4.1	936/435	0.000226
detect	advanced	`mistralai/mistral-large-2512`	4.3	936/396	0.001168

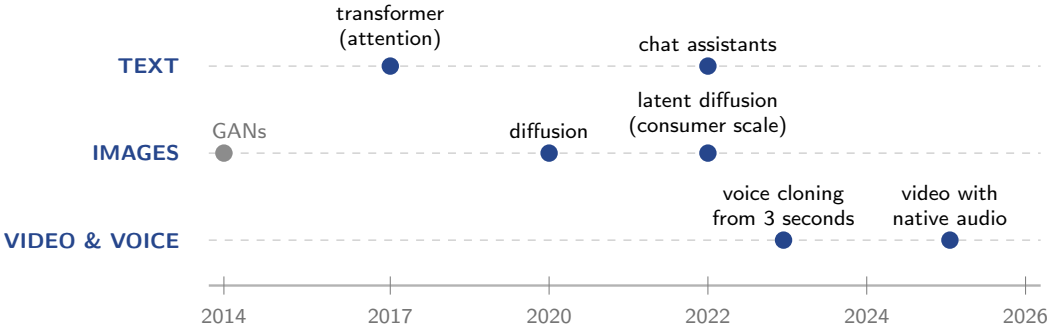
Real outputs, recorded in the app (`recorded_run.json`); reviewed during preparation; order and blind IDs fixed at freezing.

Rehearsal run in the app: the matrix as displayed

	Economy detector `mistralai/ministral-8b-2512` **RECORDED**	**Advanced detector** `mistralai/mistral-large-2512` **RECORDED**
Economy generator (recorded) `mistralai/ministral-8b-2512`	**3/3** correct of eligible (3/3 = 100%) coverage: 3/3 = 100% answered abstentions: 0 · failures: 0 (service 0, format 0) false-claim recall: 2/2 = 100% false positives: 0 of 1 true · rate 0/1 = 0% answered-only accuracy: 3/3 = 100% **imbalanced cell**	**3/3** correct of eligible (3/3 = 100%) coverage: 3/3 = 100% answered abstentions: 0 · failures: 0 (service 0, format 0) false-claim recall: 2/2 = 100% false positives: 0 of 1 true · rate 0/1 = 0% answered-only accuracy: 3/3 = 100% **imbalanced cell**
Advanced generator (recorded) `mistralai/mistral-large-2512`	**4/4** correct of eligible (4/4 = 100%) coverage: 4/4 = 100% answered abstentions: 0 · failures: 0 (service 0, format 0) false-claim recall: 2/2 = 100% false positives: 0 of 2 true · rate 0/2 = 0% answered-only accuracy: 4/4 = 100%	**4/4** correct of eligible (4/4 = 100%) coverage: 4/4 = 100% answered abstentions: 0 · failures: 0 (service 0, format 0) false-claim recall: 2/2 = 100% false positives: 0 of 2 true · rate 0/2 = 0% answered-only accuracy: 4/4 = 100%

Screenshot of the prepared full app (`fallback/full-app`, Detect & compare tab), recorded rehearsal run of 18 Oct 2026, real model outputs.

From the source deck: three families — one architecture underneath



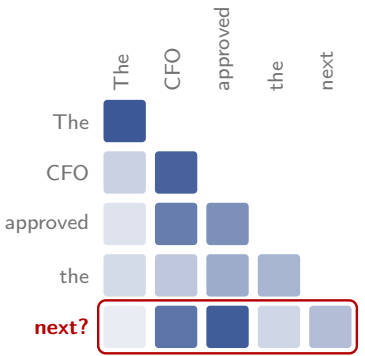
The lanes have converged

The 2017 text architecture — the *transformer* — now sits inside most new image and video systems too.

Source deck p. 9. Vaswani et al. (2017); Goodfellow et al. (2014); Ho, Jain, and Abbeel (2020); Rombach et al. (2022); Chen et al. (2025).

From the source deck: one level deeper — attention

- **Embeddings:** each token becomes a vector — similar meanings end up near each other
- **Self-attention:** every position weighs every earlier one — "which words matter here?" — all in parallel, which is why it scales
- **Stack + scale:** dozens of such layers, billions of parameters; capability emerges largely from scale
- **Why it matters:** attention gives the model *global context* — the ingredient of long, coherent, stylistically consistent fakes



The next token leans on "CFO" and "approved".

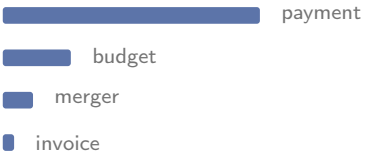
Darker = more attention. Illustrative weights.

Source deck p. 13. Vaswani et al. (2017); attention itself: Bahdanau, Cho, and Bengio (2015).

From the source deck: why the same prompt never gives the same answer

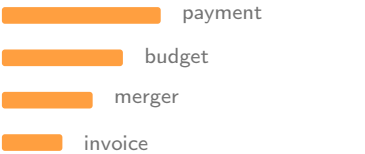
The model outputs a distribution — something still has to *pick*.

Low "temperature" — safe



predictable, repetitive — "corporate" prose

High "temperature" — creative



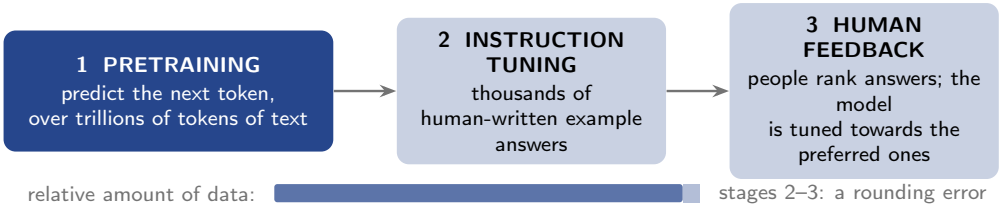
surprising, varied — and more likely to wander

Governance consequence

Output is not reproducible — ask twice, get two answers. Audit the *logged* output; you may not be able to re-run it.

Source deck p. 14. Temperature scales the scores before they become probabilities; see Holtzman et al. (2020).

From the source deck: from raw predictor to assistant — three stages



- Stage 1 buys the *capability* — the knowledge, the style, the fluency
- Stages 2–3 buy the *behaviour* — helpfulness, tone, refusals

- **So guardrails are a thin layer over a very general engine — and "we fine-tuned it" is not a safety strategy**

Source deck p. 15. The classic recipe: Ouyang et al. (2022). Newer models add further reinforcement-learning stages.

From the source deck: your turn (Question 1 of 3)

Question 1 — discuss or type in the chat

A plausibility engine writes your first draft.

Where in *your* studies or work would that be genuinely useful — and where would it be quietly dangerous?

Name one of each: **useful:** . . . / **risky:** . . .

No wrong answers — the next slide shows what the pattern usually is.

Source deck p. 17 (optional in this lecture).

From the source deck: the pattern behind your answers

✔ Where it is genuinely useful

Drafting, summarising, brainstorming, translating, code you are going to run anyway — everything where *a human already checks the output before it matters*.

⚠ Where it is quietly dangerous

Citations, numbers, legal or medical claims, anything a customer sees, anything you sign — everything that *ships without a verification step*.

The rule worth writing down

The risk is not the model — it is the *missing step* between generation and consequence. The same output is excellent in a draft and dangerous in a filing. **So "don't use AI" is the wrong policy — "never ship unverified output" is the right one.**

Source deck p. 18. Fabricated citations in court: U.S. District Court, S.D.N.Y. (2023).

Image credits & licences

- **Synthetic face (Round 1A):** "Woman 1.jpg" — StyleGAN2 / thispersondoesnotexist.com output, via Wikimedia Commons, public domain (PD-algorithm).
- **Real portrait (Round 1B):** "Headshot C. Estelle Smith" by FauxNeme (CC BY-SA 4.0), via Wikimedia Commons; cropped.
- **Arc de Triomphe pair (Round 2):** real: "Arc de Triomphe de l'Étoile at night" by Wilfredor (CC0); AI: "Christmas at Arc de Triomphe 2023" by Sérgio Valle Duarte via Midjourney (CC BY 3.0), both via Wikimedia Commons.
- **Presenter photograph:** from the author's chapter-00 course deck.
- **App screenshots:** the prepared full app (`fallback/full-app`), recorded rehearsal run of 18 Oct 2026.
- **Everything else:** original TikZ diagrams, adapted from the source deck.

References I

U.S. District Court, S.D.N.Y. (2023). *Mata v. Avianca, Inc., 678 F. Supp. 3d 443 (S.D.N.Y.)* No. 22-cv-1461 (PKC), Opinion and Order on Sanctions, 22 June 2023: $5,000 penalty on two attorneys and their firm for non-existent, ChatGPT-generated opinions.

Bahdanau, D., K. Cho, and Y. Bengio (2015). "Neural Machine Translation by Jointly Learning to Align and Translate". In: *International Conference on Learning Representations (ICLR)*. The attention mechanism itself, two years before the transformer made it the whole architecture. arXiv: 1409.0473.

Brown, T. B., B. Mann, N. Ryder, M. Subbiah, et al. (2020). "Language Models are Few-Shot Learners". In: *Advances in Neural Information Processing Systems 33 (NeurIPS 2020)*. Vol. 33, pp. 1877–1901. arXiv: 2005.14165.

Buffett, W. E. (2014). *Letter to the shareholders of Berkshire Hathaway Inc. for 2013*. Berkshire Hathaway Inc., 2013 Annual Report; letter dated 28 February 2014; berkshirehathaway.com/letters/2013ltr.pdf. URL: https://www.berkshirehathaway.com/letters/2013ltr.pdf.

Chen, S. et al. (2025). "Neural Codec Language Models are Zero-Shot Text to Speech Synthesizers". In: *IEEE Transactions on Audio, Speech and Language Processing* 33, pp. 705–718. DOI: 10.1109/TASLPRO.2025.3530270. arXiv: 2301.02111. VALL-E; 2023 preprint by C. Wang, S. Chen et al.: speech from "only a 3-second enrolled recording".

DeepSeek-AI (2025). "DeepSeek-R1 incentivizes reasoning in LLMs through reinforcement learning". In: *Nature* 645, pp. 633–638.

Goodfellow, I. J. et al. (2014). "Generative Adversarial Nets". In: *Advances in Neural Information Processing Systems 27 (NIPS 2014)*. Vol. 27, pp. 2672–2680. arXiv: 1406.2661. URL: https://papers.nips.cc/paper_files/paper/2014/hash/f033ed80deb0234979a61f95710dbe25-Abstract.html.

Ho, J., A. Jain, and P. Abbeel (2020). "Denoising Diffusion Probabilistic Models". In: *Advances in Neural Information Processing Systems 33 (NeurIPS 2020)*. Vol. 33, pp. 6840–6851. arXiv: 2006.11239. URL: https://proceedings.neurips.cc/paper/2020/hash/4c5bcfec8584af0d967f1ab10179ca4b-Abstract.html.

References II

Holtzman, A. et al. (2020). "The Curious Case of Neural Text Degeneration". In: *International Conference on Learning Representations (ICLR)*. Canonical reference for nucleus (top-p) sampling and for the comparison of decoding strategies. arXiv: 1904.09751. URL: https://openreview.net/forum?id=rygGQyrFvH.

Jakesch, M., J. T. Hancock, and M. Naaman (2023). "Human heuristics for AI-generated language are flawed". In: *Proceedings of the National Academy of Sciences* 120.11, e2208839120. DOI: 10.1073/pnas.2208839120.

Kalai, A. T., O. Nachum, S. S. Vempala, and E. Zhang (2025). *Why Language Models Hallucinate*. arXiv: 2509.04664.

Miller, E. J. et al. (2023). "AI hyperrealism: Why AI faces are perceived as more real than human ones". In: *Psychological Science* 34.12, pp. 1390–1403. DOI: 10.1177/09567976231207095.

Mistral AI (2025a). *Ministral 3 8B Instruct 2512 — model card*. 8.4B-parameter language model plus a 0.4B vision encoder. URL: https://huggingface.co/mistralai/Ministral-3-8B-Instruct-2512.

— (2025b). *Mistral Large 3 675B Instruct 2512 — model card*. Mixture-of-experts with 41B active and 675B total parameters. URL: https://huggingface.co/mistralai/Mistral-Large-3-675B-Instruct-2512.

NewsGuard (Feb. 2026). *Quarterly AI False Claim Monitor — January 2026*. 11 leading chatbots tested on 10 false news claims: "more than 28%" repeat rate on controversial topics. NewsGuard Technologies. URL: https://www.newsguardtech.com/ai-monitor/january-2026/.

Nightingale, S. J. and H. Farid (2022). "AI-synthesized faces are indistinguishable from real faces and more trustworthy". In: *Proceedings of the National Academy of Sciences* 119.8, e2120481119. DOI: 10.1073/pnas.2120481119.

OpenRouter (2026). *API reference: chat completions, models, structured outputs, provider routing, errors*. URL: https://openrouter.ai/docs/quickstart.

Ouyang, L. et al. (2022). "Training language models to follow instructions with human feedback". In: *Advances in Neural Information Processing Systems 35 (NeurIPS 2022)*. Vol. 35, pp. 27730–27744. DOI: 10.52202/068431-2011. arXiv: 2203.02155.

References III

Petrov, A., E. La Malfa, P. H. S. Torr, and A. Bibi (2023). "Language Model Tokenizers Introduce Unfairness Between Languages". In: *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*. Vol. 36, pp. 36963–36990. DOI: 10.52202/075280-1608. arXiv: 2305.15425. URL: https://papers.nips.cc/paper_files/paper/2023/hash/74bb24dca8334adce292883b4b651eda-Abstract-Conference.html.

Rombach, R. et al. (2022). "High-Resolution Image Synthesis With Latent Diffusion Models". In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 10684–10695. arXiv: 2112.10752. URL: https://openaccess.thecvf.com/content/CVPR2022/html/Rombach_High-Resolution_Image_Synthesis_With_Latent_Diffusion_Models_CVPR_2022_paper.html.

Snowflake Inc. (2026). *Streamlit documentation: session state, forms, secrets, AppTest*. URL: https://docs.streamlit.io/.

Vaswani, A. et al. (2017). "Attention is All you Need". In: *Advances in Neural Information Processing Systems 30 (NIPS 2017)*. Vol. 30, pp. 5998–6008. arXiv: 1706.03762. URL: https://papers.nips.cc/paper_files/paper/2017/hash/3f5ee243547dee91fbd053c1c4a845aa-Abstract.html.

World Health Organization (June 2026). *Tobacco and nicotine*. Fact sheet. Source of the evidence packet E1–E4 and of the correction shown in the app. World Health Organization. URL: https://www.who.int/news-room/fact-sheets/detail/tobacco.

Resources and contact

Course materials from Hochschule Bielefeld (HSBI)

AI or Reality? (this lecture) slides, prompts, tested Streamlit app
https://github.com/ChrisW09/AI-or-Reality

Python for AI-Driven Automation Python, AI-powered automation, agentic AI
https://github.com/ChrisW09/Python-for-AI-Driven-Automation

Quantitative Research Methods statistical methods, machine learning, quantitative analysis
https://github.com/ChrisW09/Quantitative-Research-Methods

Contact

Prof. Dr. Christoph Weisser — Hochschule Bielefeld (HSBI) christoph.weisser@hsbi.de

Thank you — questions welcome.